Physical chemistry

Mark scheme and worked solutions for 25 questions: 8 multiple choice, 17 structured.

1 · Part 1

Paper 1 & 2 | Energetics | [1]

Answer: B

Worked solution

An endothermic reaction takes in heat energy from the surroundings. Calcium carbonate only breaks down into calcium oxide and carbon dioxide while it is being strongly heated, because the reaction needs that energy. Burning a fuel, neutralising an acid and reacting sodium with water all make their surroundings hotter, so they are exothermic.

2 · Part 1

Paper 1 & 2 | Energetics | [1]

Answer: D

Worked solution

Temperature change $\Delta T = 34.0 - 18.0 = 16.0\ °C$. $Q = m \times c \times \Delta T = 50\ g \times 4.2\ J/g/°C \times 16.0\ °C = 3360\ J$. Divide by 1000 to convert to kilojoules: 3.36 kJ.

3 · Part 1

Paper 2 only | Energetics | [1]

Answer: A

Worked solution

Breaking a bond always needs energy to be taken in (endothermic). Making a bond always releases energy (exothermic). Only change A forms a new bond without breaking any. Changing a liquid into a gas also takes in energy, so C is not exothermic.

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4 · Part 1 Paper 1 & 2 Energetics [7]

Mark scheme

(a) **[4]**

Indicative points, any four (1 each):

- measure a known volume of copper(II) sulfate solution (e.g. 25 cm^3) with the measuring cylinder and pour it into the polystyrene cup (1)
- measure the starting temperature of the solution with the thermometer (1)
- weigh out zinc powder on the balance and add it to the solution (1)

ALLOW add an excess of zinc powder

- put the lid on and stir the mixture (1)
- record the highest temperature reached (1)
- subtract the starting temperature from the highest temperature to find the temperature rise (1)

(b) **[1]**

- polystyrene is a (thermal) insulator OR the lid and cup reduce heat loss to the surroundings (1)

ALLOW keeps the heat in

IGNORE "it is safer"

(c) **[2]**

- powder has a larger surface area so the reaction is faster (1)
- so the highest temperature is reached quickly, before much heat is lost to the surroundings (1)

ALLOW all the zinc reacts quickly OR less time for heat to escape

Worked solution

(a) The aim is to measure how much the temperature of the solution goes up. Measure a fixed volume of copper(II) sulfate solution into the polystyrene cup and take its starting temperature. Weigh the zinc powder, add it, put on the lid and stir so the heat spreads through the solution. Watch the thermometer and record the highest temperature. The temperature rise is the highest temperature minus the starting temperature.

(b) Polystyrene is a poor conductor of heat and the lid stops heat escaping from the surface, so less heat energy is lost to the air and the temperature rise is closer to the true value.

(c) Zinc powder has a much larger surface area than one lump, so the reaction finishes quickly. If the reaction were slow, heat would be escaping from the cup the whole time, so the highest temperature recorded would be too low.

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5 · Part 1 Paper 1 & 2 Energetics [7]

Mark scheme

(a) **[2]**

- temperature rise = 32.5 − 19.0 = 13.5 °C and Q = 50.0 × 4.2 × 13.5 (1)
- Q = 2835 J (1)

ALLOW 2840 J OR 2800 J

(b) **[3]**

- Q in kJ = 2835 ÷ 1000 = 2.835 kJ (1)
- 2.835 ÷ 0.0500 = 56.7 (1)
- ΔH = −56.7 kJ/mol, with negative sign (1)

ALLOW −57 kJ/mol OR −56.8 kJ/mol

ECF from the value of Q in (a)

(c) **[2]**

- temperature rise is halved OR about 6.8 °C (1)

ALLOW 6.75 °C OR 6.7 °C

- the same amount of acid and alkali react so the same heat energy is released, but it heats twice the mass of mixture (1)

Worked solution

(a) ΔT = 32.5 − 19.0 = 13.5 °C. Q = m × c × ΔT = 50.0 × 4.2 × 13.5 = 2835 J.

(b) Change Q into kilojoules: 2835 J ÷ 1000 = 2.835 kJ. Divide by the amount of HCl that reacted: 2.835 kJ ÷ 0.0500 mol = 56.7 kJ/mol. The temperature went up, so the reaction is exothermic and ΔH is negative: ΔH = −56.7 kJ/mol.

(c) The amounts of HCl and NaOH are the same, so the same amount of water forms and the same heat energy, 2835 J, is released. This energy now heats 100 g instead of 50 g. Rearranging Q = mcΔT gives ΔT = 2835 ÷ (100 × 4.2) = 6.75 °C, which is half the original rise.

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6 · Part 1

Paper 1 & 2 | Energetics | [9]

Mark scheme

(a) **[3]**

- mass of hexane burned = 152.37 − 151.51 = 0.86 g (1)
- Mr of C_6H_{14} = (6 × 12) + (14 × 1) = 86 (1)
- amount = 0.86 ÷ 86 = 0.0100 mol (1)

ALLOW 0.01 mol

ECF from an incorrect mass or Mr

(b) **[2]**

- temperature rise = 43.5 − 18.5 = 25.0 °C and Q = 200 × 4.2 × 25.0 = 21 000 J (1)
- Q = 21.0 kJ (1)

ALLOW 21 kJ

(c) **[2]**

- 21.0 ÷ 0.0100 = 2100 (1)
- ΔH = −2100 kJ/mol, with negative sign (1)

ECF from (a) and (b)

(d) **[2]**

- heat energy is lost to the surroundings OR heats the air OR heats the copper can (1)
- so less heat energy goes into the water OR the temperature rise is smaller, so the calculated Q is too small (1)

ALLOW incomplete combustion of hexane (1) so less energy is released than in complete combustion (1)

ALLOW some hexane evaporates from the wick without burning (1) so the mass loss is too large for the energy measured (1)

Worked solution

(a) The mass of hexane burned is the drop in mass of the burner: 152.37 − 151.51 = 0.86 g. Using the Periodic Table, Mr of C_6H_{14} = 6 × 12 + 14 × 1 = 86. Amount = mass ÷ Mr = 0.86 ÷ 86 = 0.0100 mol.

(b) ΔT = 43.5 − 18.5 = 25.0 °C. Q = 200 × 4.2 × 25.0 = 21 000 J = 21.0 kJ.

(c) ΔH = −Q ÷ amount = −21.0 ÷ 0.0100 = −2100 kJ/mol. It is negative because combustion is exothermic.

(d) Much of the heat from the flame does not reach the water: it warms the air around the can and the can itself. The water therefore gets less energy, the temperature rise is too small and the calculated value of ΔH is less negative than the true value. Incomplete combustion also releases less energy than complete combustion.

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7 · Part 1 Paper 1 & 2 Energetics [8]

Mark scheme

(a) **[2]**

- W −6.5, X +7.5, Y −2.0, Z +12.5
- all four correct with correct signs (2)
- two or three correct (1)

ALLOW −6.5 OR -6.5 (hyphen as minus); ALLOW: 7.5 and 12.5 without a + sign

(b) **[2]**

- W and Y (both needed) (1)
- the temperature decreases OR heat energy is taken in from the water (1)

(c) **[2]**

- Z (1)
- largest temperature rise OR gives out the most heat energy (1)

(d) **[2]**

Any two (1 each):

- mass of salt (5.0 g) (1)
- volume of water (50 cm^3) (1)
- type of cup OR insulation OR use of a lid (1)
- amount of stirring (1)

IGNORE "the same amount" without saying what

Worked solution

(a) Temperature change = final temperature − starting temperature. W: 14.0 − 20.5 = −6.5 °C. X: 27.5 − 20.0 = +7.5 °C. Y: 19.0 − 21.0 = −2.0 °C. Z: 33.0 − 20.5 = +12.5 °C.

(b) When a substance dissolves endothermically it takes in heat energy from the water, so the temperature falls. W and Y both show a fall.

(c) A self-heating can needs a change that gives out as much heat as possible. Z gives the largest temperature rise.

(d) Only the type of salt should change. Chen kept the mass of salt and the volume of water the same, and also used the same type of cup and stirred each mixture. (His starting temperatures were not all the same, so that is not a controlled variable here.)

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8 · Part 1 Paper 2 only Energetics [6]

Mark scheme

(a) **[4]**

- energy to break bonds = (4 × 391) + 158 + 498 = 2220 kJ (1)
- energy released making bonds = 945 + (4 × 464) = 2801 kJ (1)
- ΔH = 2220 − 2801 (1)
- ΔH = −581 kJ/mol, with negative sign (1)

ECF from incorrect totals for bonds broken or made

(b) **[2]**

- breaking bonds takes in energy (endothermic) and making bonds gives out energy (exothermic) (1)
- more energy is given out when the bonds in the products are made than is taken in when the bonds in the reactants are broken (1)

Worked solution

(a) Bonds broken (reactants): one N_2H_4 molecule has 4 N–H bonds and 1 N–N bond, and one O_2 has 1 O=O bond. Energy taken in = 4 × 391 + 158 + 498 = 1564 + 158 + 498 = 2220 kJ. Bonds made (products): one N≡N bond and two H_2O molecules with 2 O–H bonds each, so 4 O–H bonds. Energy given out = 945 + 4 × 464 = 945 + 1856 = 2801 kJ. ΔH = bonds broken − bonds made = 2220 − 2801 = −581 kJ/mol.

(b) Bond breaking is endothermic and bond making is exothermic. Here the energy released by making the new bonds (2801 kJ) is greater than the energy needed to break the old bonds (2220 kJ), so there is an overall release of heat energy.

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9 · Part 1 Paper 2 only Energetics [6]

Mark scheme

(a) **[3]**

- energy to break bonds = 436 + 193 = 629 kJ (1)
- 629 − (2 × H–Br) = −103 OR energy released making bonds = 629 + 103 = 732 kJ (1)
- H–Br bond energy = 732 ÷ 2 = 366 kJ/mol (1)

ECF from an incorrect total for bonds broken

(b) **[3]**

- bonds broken = 436 + 158 = 594 kJ and bonds made = 2 × 568 = 1136 kJ (1)
- ΔH = 594 − 1136 = −542 kJ/mol, with negative sign (1)
- exothermic, because ΔH is negative OR more energy is released making bonds than is taken in breaking bonds (1)

ECF deduction consistent with the sign of their ΔH

Worked solution

(a) Energy taken in to break one H–H bond and one Br–Br bond = 436 + 193 = 629 kJ. ΔH = bonds broken − bonds made, so −103 = 629 − bonds made, giving bonds made = 732 kJ. Two H–Br bonds are made, so each H–Br bond has a bond energy of 732 ÷ 2 = 366 kJ/mol. Check: 629 − 732 = −103 kJ/mol.

(b) Bonds broken: one H–H and one F–F = 436 + 158 = 594 kJ. Bonds made: two H–F = 2 × 568 = 1136 kJ. ΔH = 594 − 1136 = −542 kJ/mol. The value is negative, so the reaction is exothermic.

10 · Part 1 Paper 1 & 2 Rates of reaction [1]

Answer: C

Worked solution

Only the zinc particles on the surface of the solid can be hit by acid particles. Breaking the zinc into powder exposes many more particles, so there are more collisions per second and the reaction is faster.

11 · Part 1 Paper 1 & 2 Rates of reaction [1]

Answer: A

Worked solution

Because a catalyst is chemically unchanged, it can be recovered at the end and used again. It only makes the reaction happen faster; the same amount of product forms as without it.

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12 · Part 1 Paper 1 & 2 Rates of reaction [1]

Answer: B

Worked solution

With the acid in excess, all the marble reacts, so the total volume of gas depends only on the mass of marble. Changing to powder keeps the mass the same but increases the surface area, so the rate increases while the total volume stays the same. Doubling the mass would also double the total volume of gas, so it does not fit the question.

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13 · Part 1 Paper 1 & 2 Rates of reaction [8]

Mark scheme

(a) **[1]**

- carbon dioxide (gas) escapes from the flask (1)

ALLOW a gas is given off and leaves the flask

(b) **[2]**

- 0.74 ÷ 60 (1)
- 0.012 g/s (1)

ALLOW 0.0123 g/s

(c) **[2]**

- the concentration of the acid decreases OR there are fewer acid particles in a given volume (as they are used up) (1)
- so there are fewer collisions per second (between acid particles and the marble) (1)

ALLOW collisions are less frequent

REJECT fewer collisions (without reference to time) for the second mark

(d) **[1]**

- all the hydrochloric acid has been used up OR has reacted (1)

REJECT all the marble has been used up

(e) **[2]**

- use (at least three) different concentrations of hydrochloric acid (1)
- keep the same volume of acid and the same mass and size of marble chips (1)

ALLOW keep the temperature the same as an alternative to one of the other controls for the second mark

Worked solution

(a) One of the products is carbon dioxide gas. It escapes through the cotton wool, so the mass on the balance falls.

(b) Mean rate = loss in mass ÷ time = 0.74 g ÷ 60 s = 0.012 g/s.

(c) Acid particles are used up as the reaction goes on, so the concentration of the acid falls. There are fewer acid particles in each cm^3, so collisions with the marble happen less often and the rate decreases.

(d) The marble is in excess, so the reaction stops when all the acid has reacted. After that no more carbon dioxide is made and the mass stays constant.

(e) Change only the concentration of the acid: repeat the experiment with several different concentrations. Keep the volume of acid, the mass and size of the marble chips and the temperature the same, and record the mass loss at the same times.

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14 · Part 1

Paper 1 & 2 | Rates of reaction | [8]

Mark scheme

(a) **[2]**

- $2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$
- correct balancing numbers, 2 and 2 (1)
- correct state symbols, (l) for water and (g) for oxygen (1)

(b) **[2]**

- manganese(IV) oxide (1)
- it produced the largest volume of oxygen in 60 s OR it gave the fastest rate (1)

(c) **[1]**

- sand, because it produced the same volume of oxygen (1 cm^3) as with no solid (1)
- both the solid and the reason needed

(d) **[3]**

- weigh the manganese(IV) oxide before adding it (1)
- filter the mixture at the end to collect the solid (1)
- wash, dry and reweigh the solid, and show its mass is the same as at the start (1)

ALLOW dry and reweigh without "wash"

Worked solution

(a) Hydrogen peroxide breaks down into water and oxygen. Two H_2O_2 molecules contain 4 H and 4 O atoms, which make 2 H_2O and 1 O_2. Water is a liquid (l) and oxygen is a gas (g).

(b) The solid that makes the most oxygen in the same time gives the fastest rate. Manganese(IV) oxide gave 52 cm^3, far more than any other solid.

(c) Sand gave 1 cm^3, the same as with no solid at all, so it does not speed up the reaction.

(d) A catalyst is chemically unchanged at the end of the reaction, so its mass should be the same. Weigh the manganese(IV) oxide first. After the reaction, filter off the solid, wash it, dry it and weigh it again. The mass should be the same as before.

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15 · Part 1 Paper 1 & 2 Rates of reaction [6]

Mark scheme

(a) **[2]**

- at higher pressure there are more gas particles in a given volume OR the particles are closer together (1)
- so there are more frequent collisions OR more collisions per second (1)

(b) **[3]**

- particles have more (kinetic) energy OR move faster (1)
- so they collide more often OR more collisions per second (1)
- a greater proportion of collisions have energy equal to or greater than the activation energy OR more collisions are successful (1)

(c) **[1]**

- the rate decreases (1)

ALLOW the reaction is slower

Worked solution

(a) Increasing the pressure squeezes the same gas into a smaller space, so there are more particles in each unit of volume. The particles collide more often, so the rate increases.

(b) At a higher temperature the particles move faster, so they collide more often. They also have more energy, so a greater proportion of the collisions have at least the activation energy and lead to a reaction. Both effects increase the rate.

(c) In a larger container the gas particles are more spread out (the pressure is lower), so they collide less often and the rate decreases.

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16 · Part 1 Paper 1 & 2 Rates of reaction [7]

Mark scheme

(a) **[4]**

Indicative points, any four (1 each):

- heat the sodium thiosulfate solution (or both solutions) to the required temperature, e.g. in a water bath (1)
- check the temperature with a thermometer (1)
- add the hydrochloric acid to the flask on the cross and start the timer at the same time (1)
- stop the timer when the cross can no longer be seen from above (1)
- use the same volumes and concentrations of both solutions each time (1)
- repeat at each of the five temperatures (1)

(b) **[2]**

- as the temperature increases, the time decreases so the rate increases (1)
- every 10 °C rise roughly halves the time OR roughly doubles the rate, with data quoted e.g. 124 s at 20 °C and 64 s at 30 °C (1)

(c) **[1]**

- the time is very short, so the reaction time of the person starting and stopping the timer has a large effect OR it is hard to judge exactly when the cross disappears in a short time (1)

ALLOW the mixture cools quickly from 60 °C during the reaction

ALLOW a noticeable amount of reaction happens while mixing, before timing starts

Worked solution

(a) Warm the sodium thiosulfate solution to the chosen temperature in a water bath and check it with a thermometer. Put the flask on the cross, add the acid and start the timer straight away. Look down through the mixture and stop the timer when the cross can no longer be seen. Keep the volumes and concentrations of the solutions the same and repeat at each temperature.

(b) The time gets shorter as the temperature rises, so the reaction gets faster. For each 10 °C rise the time roughly halves (124 s, 64 s, 33 s, 17 s, 9 s), so the rate roughly doubles.

(c) At 60 °C the cross disappears in only 9 s. A delay of even half a second in starting or stopping the timer is a large part of 9 s, so this result has the largest percentage error.

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17 · Part 1

Paper 1 & 2 | Rates of reaction | [6]

Mark scheme

(a) **[2]**

- 38 ÷ 20 (1)
- 1.9 cm^3/s (1)

(b) **[2]**

- gradient = (79 − 37) ÷ (70 − 10) = 42 ÷ 60 (1)
- 0.70 cm^3/s, with unit (1)

ALLOW 0.7 cm^3/s

(c) **[2]**

- 72 cm^3 OR the same final volume (1)
- the same amount of magnesium reacts (magnesium is the limiting reactant) OR the acid is in excess in both experiments (1)

REJECT a larger volume because the acid is more concentrated

Worked solution

(a) Mean rate = volume of gas ÷ time = 38 cm^3 ÷ 20 s = 1.9 cm^3/s.

(b) The rate at an instant is the gradient of the tangent at that time. Gradient = change in volume ÷ change in time = (79 − 37) ÷ (70 − 10) = 42 ÷ 60 = 0.70 cm^3/s. This is lower than the mean rate in the first 20 s because the reaction slows down.

(c) The acid is in excess, so the reaction stops when all of the magnesium has reacted. The piece of magnesium is identical, so the same volume of hydrogen, 72 cm^3, is made. More concentrated acid only makes the gas come off faster.

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18 · Part 1 Paper 1 & 2 Rates of reaction [5]

Mark scheme

(a) **[1]**

- the minimum energy that colliding particles need for a reaction to happen (1)

ALLOW the minimum energy needed to start a reaction

(b) **[3]**

- the catalyst provides an alternative pathway OR route for the reaction (1)
- with a lower activation energy (1)
- so a greater proportion of collisions have enough energy to react OR more successful collisions per second (1)

(c) **[1]**

- the process can be run at a lower temperature, so less energy OR fuel is needed (1)

ALLOW more product is made per hour OR the catalyst can be used again

REJECT the catalyst increases the yield

Worked solution

(a) Activation energy is the minimum amount of energy that particles must have when they collide for the collision to cause a reaction.

(b) Platinum gives the reaction a different route with a lower activation energy (135 kJ/mol instead of 250 kJ/mol). At the same temperature, many more of the colliding particles now have at least this energy, so more collisions are successful each second and the rate increases.

(c) With a catalyst the reaction is fast enough at a lower temperature, so less fuel is needed for heating. The catalyst is not used up, so it does not need to be bought again for every batch.

19 · Part 1 Paper 1 & 2 Reversible reactions and equilibria [1]

Answer: D

Worked solution

The reaction is reversible: $CuSO_4 \cdot 5H_2O(s) \rightleftharpoons CuSO_4(s) + 5H_2O(l)$. Heating drives it forwards (blue to white). Adding water drives it backwards (white to blue).

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20 · Part 1 Paper 2 only Reversible reactions and equilibria [1]

Answer: C

Worked solution

ΔH is negative, so the forward reaction is exothermic. Lowering the temperature moves the position of equilibrium in the exothermic direction (right). Counting gas moles: 1 CO + 2 H_2 = 3 mol on the left, 1 mol CH_3OH on the right. Raising the pressure moves the position of equilibrium towards fewer moles of gas (right).

21 · Part 1 Paper 1 & 2 Reversible reactions and equilibria [7]

Mark scheme

(a) **[1]**

- the reaction is reversible (1)

ALLOW the reaction can go in both directions

(b) **[2]**

- NH_3 and HCl (1)
- (g) for both products (1)
- $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$

ALLOW products in either order

(c) **[2]**

- the white solid at the bottom disappears (1)
- a white solid forms (reappears) near the top of the tube (1)

(d) **[2]**

- the ammonia and hydrogen chloride gases rise to the cooler part of the tube (1)
- where the reverse reaction happens OR they react together to form ammonium chloride again (1)

Worked solution

(a) The symbol ⇌ shows that the reaction is reversible: the products can react to form the reactants again.

(b) Ammonia is NH_3 and hydrogen chloride is HCl. Both are gases, so: $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$.

(c) At the hot bottom of the tube the white solid seems to disappear as it breaks down into gases. Near the cool top a white solid appears on the glass.

(d) The two gases rise up the tube. At the cooler top the reaction goes the other way: ammonia and hydrogen chloride combine to form solid ammonium chloride again.

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22 · Part 1 Paper 2 only Reversible reactions and equilibria [8]

Mark scheme

(a) **[1]**

- so that none of the gases (reactants or products) can escape (1)

ALLOW it must be a closed system OR no substance can enter or leave

(b) **[2]**

- the forward and reverse reactions take place at the same rate (1)
- so the concentration of NO_2 (and N_2O_4) stays constant (1)

ALLOW NO_2 is made as fast as it is used up

(c) **[3]**

- darker brown means there is more NO_2 (1)
- so the position of equilibrium has moved to the right (1)
- an increase in temperature shifts the equilibrium in the direction of the endothermic reaction, so the forward reaction is endothermic (1)

(d) **[2]**

- the position of equilibrium moves to the left (1)
- because there are fewer moles of gas on the left (1 mol) than on the right (2 mol) (1)

Worked solution

(a) In an open container the gases would escape, so the reverse reaction could not keep pace and equilibrium could not be set up. A sealed container keeps all the reactants and products together.

(b) At dynamic equilibrium the forward and reverse reactions are still happening, but at the same rate. NO_2 is formed exactly as fast as it is used up, so its concentration, and the colour, stays constant.

(c) The darker colour shows more brown NO_2 has formed, so the equilibrium has moved to the right. Raising the temperature shifts the equilibrium in the endothermic direction, so the forward reaction must be endothermic.

(d) Left side: 1 mol of gas (N_2O_4). Right side: 2 mol of gas (NO_2). Increasing the pressure shifts the equilibrium towards fewer moles of gas, so it moves to the left.

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23 · Part 1 Paper 2 only Reversible reactions and equilibria [7]

Mark scheme

(a) **[2]**

- there are fewer moles of gas on the right (2 mol) than on the left (3 mol) (1)
- so increasing the pressure shifts the position of equilibrium to the right and the yield increases (1)

(b) **[2]**

- the catalyst increases the rates of the forward and reverse reactions equally (by the same amount) (1)
- so the position of equilibrium does not change; equilibrium is only reached faster (1)

(c) **[3]**

- the forward reaction is exothermic, so a lower temperature shifts the equilibrium to the right (1)
- at a lower temperature the rate of reaction is too slow (1)
- 450 °C gives a reasonable yield at a fast enough rate OR is a compromise between yield and rate (1)

Worked solution

(a) There are 2 + 1 = 3 moles of gas on the left and 2 moles on the right. Increasing the pressure shifts the equilibrium towards fewer moles of gas, so more SO_3 forms.

(b) A catalyst speeds up the forward and reverse reactions by the same amount. Both rates stay equal to each other, so the position of equilibrium and the yield are unchanged. The catalyst simply lets equilibrium be reached sooner.

(c) ΔH is negative, so the forward reaction is exothermic. Lowering the temperature shifts the equilibrium in the exothermic direction, to the right, giving more SO_3. However, at a low temperature the reaction is very slow, so little product would be made each hour. About 450 °C is a compromise that gives a reasonable yield quickly.

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24 · Part 1 Paper 2 only Reversible reactions and equilibria [5]

Mark scheme

(a) **[2]**

- endothermic (1)
- the percentage of methane converted increases as the temperature increases, and an increase in temperature shifts the equilibrium in the direction of the endothermic reaction (1)

ALLOW data quoted, e.g. 30% at 600 °C rising to 93% at 900 °C

(b) **[2]**

- there are fewer moles of gas on the left (2 mol) than on the right (4 mol) (1)
- increasing the pressure shifts the position of equilibrium to the left (towards fewer moles of gas), so less methane is converted (1)

(c) **[1]**

- a higher pressure gives a faster rate of reaction OR more product is made in a given time (1)

ALLOW smaller equipment can be used OR the gases are already at high pressure for later use

Worked solution

(a) At 10 atm the percentage converted rises from 30% at 600 °C to 93% at 900 °C. A higher temperature shifts the equilibrium in the endothermic direction. The equilibrium moves to the right as temperature rises, so the forward reaction is endothermic.

(b) Left side: 1 CH_4 + 1 H_2O = 2 mol of gas. Right side: 1 CO + 3 H_2 = 4 mol of gas. Increasing the pressure shifts the equilibrium towards fewer moles of gas, which is to the left, so less methane is converted.

(c) At higher pressure the gas particles are closer together and collide more often, so the reaction is faster and more hydrogen is made each hour, even though the equilibrium percentage is lower.

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25 · Part 1 Paper 2 only Reversible reactions and equilibria [6]

Mark scheme

(a) **[2]**

- 4 min (1)
- the amounts of H_2, I_2 and HI stay constant from 4 min onwards (1)

(b) **[4]**

Indicative points, any four (1 each):

- at the start the forward reaction is at its fastest because the amounts (concentrations) of H_2 and I_2 are highest (1)
- at the start the rate of the reverse reaction is zero because there is no HI (1)
- the forward rate decreases as H_2 and I_2 are used up (1)
- the reverse rate increases as the amount of HI increases (1)
- at equilibrium the forward and reverse rates are equal, so the amounts stop changing (1)

Worked solution

(a) From 4 min onwards the amounts are 0.22 mol H_2, 0.22 mol I_2 and 1.56 mol HI and they do not change, so the mixture is at dynamic equilibrium.

(b) At 0 min there is plenty of H_2 and I_2, so the forward reaction is fast, but there is no HI, so the reverse reaction cannot happen. As time goes on, H_2 and I_2 are used up, so their concentrations fall and the forward reaction slows down. At the same time HI builds up, so the reverse reaction speeds up. Eventually the two rates become equal. From then on HI is made as fast as it breaks down, so the amounts of all three gases stay constant.

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